

Operating System Concepts (7th Ed.) — Chapter 1: Introduction

Slides 1 to 23 — Study Questions

Batch 1 – Questions 1 to 16: What Operating Systems Do

Q1. An operating system is best described as:

- a) A program that acts as an intermediary between a user of a computer and the computer hardware
- b) A program that only manages files
- c) A piece of hardware that speeds up the CPU
- d) A programming language compiler

Answer: a) A program that acts as an intermediary between a user of a computer and the computer hardware

Q2. Which of the following is NOT listed as an operating system goal?

- a) Execute user programs and make solving user problems easier
- b) Make the computer system convenient to use
- c) Use the computer hardware in an efficient manner
- d) Increase the physical size of main memory

Answer: d) Increase the physical size of main memory

Q3. Which of the following is NOT one of the four components of a computer system?

- a) Hardware
- b) Operating system
- c) Application programs
- d) Network cables

Answer: d) Network cables

Q4. In the four-component view of a computer system, which component provides the basic computing resources (CPU, memory, I/O devices)?

- a) Hardware
- b) Operating system
- c) Application programs
- d) Users

Answer: a) Hardware

Q5. Which component controls and coordinates the use of hardware among various applications and users?

- a) Hardware
- b) Operating system
- c) Application programs
- d) Users

Answer: b) Operating system

Q6. Word processors, compilers, web browsers, database systems, and video games are examples of:

- a) Hardware
- b) Operating systems
- c) Application programs
- d) Device controllers

Answer: c) Application programs

Q7. In the slides, 'users' can be:

- a) Only people
- b) People, machines, or other computers
- c) Only other computers
- d) Only application programs

Answer: b) People, machines, or other computers

Q8. When the OS is described as a 'resource allocator', it:

- a) Manages all resources and decides between conflicting requests for efficient and fair resource use
- b) Only allocates memory to the compiler
- c) Only allocates disk space
- d) Prevents users from using resources

Answer: a) Manages all resources and decides between conflicting requests for efficient and fair resource use

Q9. When the OS is described as a 'control program', it:

- a) Controls the execution of programs to prevent errors and improper use of the computer
- b) Controls only the printer
- c) Decides which application is installed
- d) Compiles user programs

Answer: a) Controls the execution of programs to prevent errors and improper use of the computer

Q10. According to the slides, the one program running at all times on the computer is called the:

- a) Bootstrap program
- b) Kernel
- c) Application program
- d) System program

Answer: b) Kernel

Q11. Everything other than the kernel is either:

- a) A system program (ships with the OS) or an application program
- b) A device driver or a trap
- c) An interrupt or a spool
- d) A cache or a buffer

Answer: a) A system program (ships with the OS) or an application program

Q12. When is the bootstrap program loaded?

- a) At power-up or reboot
- b) Only when an application crashes
- c) When a file is deleted
- d) When the printer is connected

Answer: a) At power-up or reboot

Q13. Where is the bootstrap program typically stored?

- a) RAM only
- b) ROM or EEPROM
- c) Magnetic tape
- d) The cache

Answer: b) ROM or EEPROM

Q14. Firmware is described in the slides as:

- a) A type of software etched directly into a piece of hardware
- b) A type of application program
- c) A kind of magnetic disk
- d) A user interface

Answer: a) A type of software etched directly into a piece of hardware

Q15. Which of the following is NOT a job of the bootstrap program?

- a) Initializing all aspects of the system
- b) Loading the operating system kernel
- c) Starting execution of the kernel
- d) Scheduling user applications on the printer

Answer: d) Scheduling user applications on the printer

Q16. EEPROM stands for:

- a) Electrically Erasable Programmable Read-Only Memory
- b) Extended Executable Program ROM Module
- c) Electronic Embedded Processor Memory
- d) Electrically Enabled Peripheral ROM

Answer: a) Electrically Erasable Programmable Read-Only Memory

Batch 2 – Questions 17 to 34: Computer-System Operation, Interrupts, and I/O

Q17. In a computer-system organization, CPUs and device controllers connect through a common bus that provides access to:

- a) Shared memory
- b) Only the hard disk
- c) Only the printer
- d) The bootstrap program

Answer: a) Shared memory

Q18. Concurrent execution of CPUs and devices leads to them:

- a) Competing for memory cycles
- b) Sharing the same registers
- c) Ignoring the bus
- d) Running only one at a time

Answer: a) Competing for memory cycles

Q19. Each device controller is in charge of:

- a) A particular device type
- b) All devices in the system
- c) Only the CPU
- d) Only main memory

Answer: a) A particular device type

Q20. Each device controller has a:

- a) Local buffer
- b) Kernel
- c) Bootstrap program
- d) Compiler

Answer: a) Local buffer

Q21. Who moves data between main memory and the local buffers of the device controllers?

- a) The CPU
- b) The user
- c) The compiler
- d) The bootstrap program

Answer: a) The CPU

Q22. How does a device controller inform the CPU that it has finished its operation?

- a) By causing an interrupt
- b) By shutting down the system
- c) By deleting its buffer
- d) By rebooting the CPU

Answer: a) By causing an interrupt

Q23. How can hardware trigger an interrupt?

- a) By sending a signal to the CPU through the system bus
- b) By executing a system call
- c) By running the compiler
- d) By rewriting the kernel

Answer: a) By sending a signal to the CPU through the system bus

Q24. How can software trigger an interrupt?

- a) By executing a special operation called a system call (monitor call)
- b) By sending a signal over the system bus
- c) By turning off the device controller
- d) By clearing the cache

Answer: a) By executing a special operation called a system call (monitor call)

Q25. The interrupt vector contains:

- a) The addresses of all the service routines
- b) The list of all installed applications
- c) The contents of main memory
- d) The disk sectors

Answer: a) The addresses of all the service routines

Q26. When are the addresses in the interrupt vector loaded by the OS?

- a) At the OS initialization phase
- b) Every time an interrupt occurs
- c) Only when the disk is full
- d) At shutdown

Answer: a) At the OS initialization phase

Q27. Why must the interrupt architecture save the address of the interrupted instruction?

- a) So the CPU can return to the interrupted computation after handling the interrupt
- b) So the interrupt vector can be erased
- c) To speed up the disk
- d) To free memory

Answer: a) So the CPU can return to the interrupted computation after handling the interrupt

Q28. Why are incoming interrupts disabled while another interrupt is being processed?

- a) To prevent a lost interrupt
- b) To speed up the CPU clock
- c) To save disk space
- d) To reset the device controller

Answer: a) To prevent a lost interrupt

Q29. A trap (exception) is:

- a) A software-generated interrupt caused by an error or a user program request
- b) A hardware failure of the bus
- c) A type of cache
- d) A type of storage device

Answer: a) A software-generated interrupt caused by an error or a user program request

Q30. Which of the following is an example of a trap caused by an error?

- a) Division by zero
- b) Printing a document
- c) Saving a file
- d) Opening a browser

Answer: a) Division by zero

Q31. The slides state that an operating system is:

- a) Interrupt driven
- b) Cache driven
- c) Disk driven
- d) Compiler driven

Answer: a) Interrupt driven

Q32. To preserve the state of the CPU during interrupt handling, the OS stores:

- a) The registers and the program counter
- b) Only the cache
- c) Only the disk sectors
- d) Only the kernel

Answer: a) The registers and the program counter

Q33. According to the slides, spooling is used for:

- a) Output (e.g., saving printer jobs in a buffer)
- b) Input only
- c) Booting the system
- d) Managing the cache

Answer: a) Output (e.g., saving printer jobs in a buffer)

Q34. Why is spooling needed for a printer?

- a) The printer cannot print all the jobs at the same time, so jobs are saved in a buffer
- b) The printer has no power
- c) The printer replaces the CPU
- d) The printer stores the kernel

Answer: a) The printer cannot print all the jobs at the same time, so jobs are saved in a buffer

Batch 3 – Questions 35 to 50: Device Controllers, I/O Structure, and DMA

Q35. The device controller is responsible for:

- a) Moving data between the peripheral devices it controls and its local buffer storage
- b) Compiling programs
- c) Scheduling the CPU
- d) Loading the bootstrap program

Answer: a) Moving data between the peripheral devices it controls and its local buffer storage

Q36. What does the OS have for each device controller to present a uniform interface to the rest of the OS?

- a) A device driver
- b) A bootstrap program
- c) A cache
- d) A spool

Answer: a) A device driver

Q37. To start an I/O operation, the device driver first:

- a) Loads the appropriate registers within the device controller
- b) Sends a trap to the user
- c) Flushes the cache
- d) Reboots the system

Answer: a) Loads the appropriate registers within the device controller

Q38. After the driver loads the registers, the controller:

- a) Examines the register contents to determine what action to take
- b) Erases the registers
- c) Turns off the device
- d) Loads the kernel again

Answer: a) Examines the register contents to determine what action to take

Q39. Then the controller starts the transfer of data from:

- a) The device to its local buffer
- b) The CPU to the compiler
- c) The cache to the registers
- d) The kernel to the disk

Answer: a) The device to its local buffer

Q40. In the first I/O method, control returns to the user program:

- a) Only upon I/O completion
- b) Immediately without waiting
- c) Never
- d) Only after a reboot

Answer: a) Only upon I/O completion

Q41. The first I/O method (waiting for completion) is suitable for:

- a) Small amounts of data
- b) Bulk data
- c) Only tape drives
- d) Only the cache

Answer: a) Small amounts of data

Q42. In the first I/O method, at most how many I/O requests are outstanding at a time?

- a) One
- b) Two
- c) Eight
- d) Unlimited

Answer: a) One

Q43. In the second I/O method, control returns to the user program:

- a) Without waiting for I/O completion
- b) Only upon I/O completion
- c) Only after a trap
- d) Only after shutdown

Answer: a) Without waiting for I/O completion

Q44. The second I/O method (no waiting) is suitable for:

- a) Large amounts of data (bulk data)
- b) Very small data only
- c) Registers only
- d) Nothing

Answer: a) Large amounts of data (bulk data)

Q45. The device-status table contains an entry for each I/O device indicating its:

- a) Type, address, and state
- b) Price, brand, and size
- c) Owner and password
- d) Color and shape

Answer: a) Type, address, and state

Q46. What does the OS do with the device-status table?

- a) Indexes into it to determine device status and to modify the table entry to include an interrupt
- b) Deletes it after boot
- c) Stores the kernel in it
- d) Uses it as a cache

Answer: a) Indexes into it to determine device status and to modify the table entry to include an interrupt

Q47. DMA stands for:

- a) Direct Memory Access
- b) Data Management Application
- c) Device Memory Allocation
- d) Dynamic Module Address

Answer: a) Direct Memory Access

Q48. DMA is used for:

- a) High-speed I/O devices able to transmit information at close to memory speeds
- b) Only slow keyboards
- c) Only the bootstrap program
- d) Only compilers

Answer: a) High-speed I/O devices able to transmit information at close to memory speeds

Q49. In DMA, the device controller transfers blocks of data directly from buffer storage to main memory:

- a) Without CPU intervention

- b) Only with CPU intervention for each byte
- c) Only after a reboot
- d) Only through the printer

Answer: a) Without CPU intervention

Q50. How many interrupts does DMA generate?

- a) One per block of data
- b) One per byte of data
- c) None at all
- d) One per second

Answer: a) One per block of data

Batch 4 – Questions 51 to 66: Storage Structure, Hierarchy, and Caching

Q51. What is the only large storage medium that the CPU can access directly?

- a) Main memory
- b) Magnetic tape
- c) Optical disk
- d) Secondary storage

Answer: a) Main memory

Q52. Secondary storage is described as:

- a) An extension of main memory that provides large nonvolatile storage capacity
- b) A small volatile storage
- c) The fastest storage in the system
- d) Part of the CPU

Answer: a) An extension of main memory that provides large nonvolatile storage capacity

Q53. Magnetic disks consist of:

- a) Rigid metal or glass platters covered with magnetic recording material
- b) Plastic ribbons
- c) Silicon chips only
- d) Optical lenses

Answer: a) Rigid metal or glass platters covered with magnetic recording material

Q54. A magnetic disk surface is logically divided into:

- a) Tracks, which are subdivided into sectors
- b) Sectors, which are subdivided into tracks
- c) Pages and frames
- d) Blocks and buffers

Answer: a) Tracks, which are subdivided into sectors

Q55. The disk controller determines:

- a) The logical interaction between the device and the computer
- b) Which application runs first
- c) The size of the cache
- d) The interrupt vector contents

Answer: a) The logical interaction between the device and the computer

Q56. Storage systems are organized in a hierarchy according to:

- a) Speed, cost, and volatility
- b) Color, size, and shape
- c) Brand, price, and warranty
- d) Age, weight, and location

Answer: a) Speed, cost, and volatility

Q57. Main memory can be viewed as:

- a) A last cache for secondary storage
- b) The slowest storage
- c) Non-volatile backup
- d) A device driver

Answer: a) A last cache for secondary storage

Q58. In the storage-device hierarchy figure, which level is at the very top (fastest)?

- a) Registers
- b) Cache
- c) Main memory
- d) Magnetic tapes

Answer: a) Registers

Q59. In the storage-device hierarchy figure, which level is at the very bottom?

- a) Magnetic tapes
- b) Optical disk
- c) Magnetic disk
- d) Electronic disk

Answer: a) Magnetic tapes

Q60. Which is the correct order of the hierarchy from top to bottom?

- a) Registers, cache, main memory, electronic disk, magnetic disk, optical disk, magnetic tapes
- b) Cache, registers, main memory, magnetic disk, electronic disk, magnetic tapes, optical disk
- c) Main memory, registers, cache, optical disk, magnetic disk, electronic disk, magnetic tapes
- d) Registers, main memory, cache, magnetic tapes, optical disk, magnetic disk, electronic disk

Answer: a) Registers, cache, main memory, electronic disk, magnetic disk, optical disk, magnetic tapes

Q61. Caching means:

- a) Copying information into a faster storage system
- b) Deleting old files
- c) Increasing disk size
- d) Rebooting the CPU

Answer: a) Copying information into a faster storage system

Q62. When information is needed, which storage is checked first?

- a) The faster storage (cache)
- b) The slower storage
- c) Magnetic tape
- d) Optical disk

Answer: a) The faster storage (cache)

Q63. If the required information is not in the cache, then:

- a) It is copied to the cache and used there

- b) The system crashes
- c) The CPU ignores the request
- d) The kernel is reloaded

Answer: a) It is copied to the cache and used there

Q64. Why is cache management an important design problem?

- a) The cache is smaller than the storage being cached
- b) The cache is larger than main memory
- c) The cache never changes
- d) The cache has no speed advantage

Answer: a) The cache is smaller than the storage being cached

Q65. Which two aspects of cache management are named in the slides?

- a) Cache size and replacement policy
- b) Cache color and shape
- c) Cache brand and price
- d) Cache owner and password

Answer: a) Cache size and replacement policy

Q66. Caching is performed at which levels?

- a) Hardware, operating system, and software
- b) Only hardware
- c) Only software
- d) Only the operating system

Answer: a) Hardware, operating system, and software

Batch 5 – Questions 67 to 78: Storage Performance Table and Data Consistency

Q67. According to the performance table, the typical size of registers is:

- a) Less than 1 KB
- b) More than 16 MB
- c) More than 16 GB
- d) More than 100 GB

Answer: a) Less than 1 KB

Q68. According to the performance table, cache is typically implemented using:

- a) On-chip or off-chip CMOS SRAM
- b) CMOS DRAM
- c) Magnetic disk
- d) Optical tape

Answer: a) On-chip or off-chip CMOS SRAM

Q69. Main memory is implemented using:

- a) CMOS DRAM
- b) CMOS SRAM
- c) Magnetic disk
- d) Custom multi-port memory

Answer: a) CMOS DRAM

Q70. Which storage level is managed by the compiler?

- a) Registers
- b) Cache
- c) Main memory
- d) Disk storage

Answer: a) Registers

Q71. Which storage level is managed by hardware?

- a) Cache
- b) Registers
- c) Main memory
- d) Disk storage

Answer: a) Cache

Q72. Which storage levels are managed by the operating system?

- a) Main memory and disk storage
- b) Registers and cache
- c) Only registers
- d) Only cache

Answer: a) Main memory and disk storage

Q73. According to the table, disk storage is typically backed by:

- a) CD or tape
- b) Cache
- c) Registers
- d) Main memory

Answer: a) CD or tape

Q74. Which storage level has the lowest access time according to the table?

- a) Registers
- b) Cache
- c) Main memory
- d) Disk storage

Answer: a) Registers

Q75. In a multitasking environment, why must the OS be careful about which value is used for a piece of data?

- a) More than one copy of the data may exist (e.g., changed in cache but not yet on disk), so the most recent value must be used
- b) Because there is only ever one copy of the data
- c) Because the cache is always up to date
- d) Because the disk is faster than the cache

Answer: a) More than one copy of the data may exist (e.g., changed in cache but not yet on disk), so the most recent value must be used

Q76. What must a multiprocessor environment provide so that all CPUs have the most recent value in their cache?

- a) Cache coherency (in hardware)
- b) More disks
- c) A larger kernel

d) A faster printer

Answer: a) Cache coherency (in hardware)

Q77. Cache coherency ensures that:

- a) An update to a value in one cache is reflected in all other caches
- b) Caches are erased regularly
- c) Only one CPU can run
- d) Main memory is disabled

Answer: a) An update to a value in one cache is reflected in all other caches

Q78. Why is data consistency even more complex in a distributed environment?

- a) Several copies of a datum can exist in distributed locations connected by a network, and all must be updated
- b) Because there is only one location
- c) Because there is no network
- d) Because caches do not exist

Answer: a) Several copies of a datum can exist in distributed locations connected by a network, and all must be updated

Essay Questions

Q79. Explain the four components of a computer system and the role of each one.

Answer:

Hardware: provides the basic computing resources (CPU, memory, I/O devices).

Operating system: controls and coordinates the use of the hardware among the various applications and users.

Application programs: define the ways in which the system resources are used to solve the computing problems of users (word processors, compilers, web browsers, database systems, video games).

Users: people, machines, or other computers.

Q80. Describe how an interrupt is handled, from the moment it occurs until the CPU resumes the interrupted work.

Answer:

- A device controller (hardware, via the system bus) or a program (software, via a system call/trap) signals an interrupt.
- The CPU stops what it is doing; the OS preserves the CPU state by saving the registers and the program counter (the address of the interrupted instruction).
- Control is transferred, through the interrupt vector (which holds the addresses of all service routines, loaded at OS initialization), to the appropriate interrupt service routine.
- Incoming interrupts are disabled while another is being processed, to prevent a lost interrupt.
- When the service routine finishes, the saved address is retrieved and the CPU returns to the interrupted computation.

Q81. Compare the two I/O methods (synchronous vs. asynchronous) and explain the role of DMA.

Answer:

Method 1: control returns to the user program only upon I/O completion. The CPU idles (wait instruction or wait loop), at most one I/O request is outstanding, and it is suitable for small amounts of data.

Method 2: control returns to the user program without waiting for I/O completion. It is suitable for bulk data; a system call lets the user wait for completion while the CPU does other work, and the device-status table tracks each device's type, address, and state.

DMA: used for high-speed devices; the device controller transfers blocks of data directly between its buffer and main memory without CPU intervention, generating only one interrupt per block instead of one per byte.

Q82. Explain caching and why maintaining consistency of data is a challenge in multitasking, multiprocessor, and distributed environments.

Answer:

Caching: information in use is copied temporarily from slower to faster storage. The faster storage (cache) is checked first; if the data is there it is used directly, otherwise it is copied to the cache and used there. Because the cache is smaller than the storage being cached, cache size and replacement policy are important design problems.

Multitasking: multiple copies of the same data may exist at different levels of the hierarchy, so the most recent value must always be used.

Multiprocessor: hardware must provide cache coherency so every CPU sees the latest value.

Distributed: copies may exist in several locations connected by a network, so all must be updated as soon as possible (and disconnections make this harder).

End of Study Questions — Slides 1 to 23.